

Captions for Data S1 to S6:

Data S1: MPRA summary statistics

This file provides summary statistics for all variants tested in MPRA stages 1 and 2.

Data S2: FrVars with functional genomic annotations

This file provides MPRA frVars intersected with functional genomic annotations from *post-mortem* human brain.

Data S3: Full CROP-seq processed data

Data table provides results from the pooled CRISPRi screen (CROP-seq) to validate 42 functional regulatory variants (frVars), including all sgRNAs, cell-types, and SNP-gene pairs surveyed.

Data S4: Pairwise LD with H1 sub-haplotype markers

Provides pairwise Linkage disequilibrium (LD) computed between H1 tag-SNP rs8070723 and frVars within the 17q21.31 locus. Also provides pairwise LD for H1C sub-haplotype tag-SNP rs242557 and frVars within 17q21.31 (CEU population ancestry).

Data S5: Figure Source Data

This file provides raw data and computed scores used for downstream analyses and figure construction. Data are organized by relevant figure.

Data S6: Supplemental Information - Methods

This file provides additional methods information including sequences, primers, and accession numbers.